

Agentic AI workflow using the OpenAI API

Create new project

```
(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agenticai-a\agents-portfolio-amith\agentic-ai-openai> mkdir project02

Directory: C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agenticai-a\agents-portfolio-amith\agentic-ai-openai

Mode                LastWriteTime         Length Name
----                -
d-----          30-12-2025   14:44                project02
```

```
graph TD
    A[User Goal] --> B[Agent (LLM + Logic)]
    B --> C[Planner (what to do next?)]
    C --> D[Tool (API / function / script)]
    D --> E[Observation (result)]
    E --> F[Decision loop]
```

- ✓ The model **reasons about actions**
- ✓ It **decides when to use a tool**
- ✓ It **observes tool output**
- ✓ It **loops with state**

```
agentic-ai-openai/
|
├── agents/
|   └── simple_agent.py    🖱️ Step 5.3
|
├── tools/
|   └── calculator.py      🖱️ Step 5.2
|
├── prompts/
|   └── system_prompt.py   🖱️ Step 5.1
|
├── main.py                🖱️ Entry point
├── .env
└── requirements.txt
```

```
(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agenticai-aiaagents-portfolio-amith\agentic-ai-openai\project02> mkdir agents

Directory: C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agenticai-aiaagents-portfolio-amith\agentic-ai-openai\project02

Mode                LastWriteTime         Length Name
----                -
d-----          30-12-2025   14:44             agents

(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agenticai-aiaagents-portfolio-amith\agentic-ai-openai\project02> mkdir tools

Directory: C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agenticai-aiaagents-portfolio-amith\agentic-ai-openai\project02

Mode                LastWriteTime         Length Name
----                -
d-----          30-12-2025   14:44             tools

(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agenticai-aiaagents-portfolio-amith\agentic-ai-openai\project02> mkdir prompts
```

```
iniconfig      2.3.0
itsdangerous   2.2.0
Jinja2         3.1.6
jiter          0.12.0
MarkupSafe     3.0.3
openai         2.14.0
packaging      25.0
pip            25.3
pluggy        1.6.0
pydantic       2.12.5
pydantic_core  2.41.5
Pygments      2.19.2
pytest         9.0.2
python-dotenv  1.2.1
requests       2.31.0
```

```
(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agentical-aiagents-portfolio-amith\agentic-ai-openai\project02> ls
```

```
Directory: C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agentical-aiagents-portfolio-amith\agentic-ai-openai\project02
```

Mode	LastWriteTime	Length	Name
d----	30-12-2025 14:45		agents
d----	30-12-2025 14:47		prompts
d----	30-12-2025 14:46		tools
-a----	30-12-2025 14:51	24	.env

```
(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\agentical-aiagents-portfolio-amith\agentic-ai-openai\project02> pip show python-dotenv
```

Name: python-dotenv

Version: 1.2.1

Summary: Read key-value pairs from a .env file and set them as environment variables

Home-page:

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License-Expression: BSD-3-Clause

Location: c:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS\.venv\Lib\site-packages

Agent step 1:

1. **Understand the goal**: The objective is to calculate the sum of 2 + 2 and provide an explanation of the result.

2. **Decide the next action**: I will perform the calculation of 2 + 2.

3. **Use available tools**: I'll do the calculation directly.

Calculation:

2 + 2 = 4

4. **Observe results**: The result of the calculation is 4.

5. **Continue until goal is achieved**: Now, I will explain the result.

Explanation: The operation performed is addition, which is one of the basic arithmetic operations. When we add the number 2 to itself, we are essentially counting two units together with another two units. This results in a total of four units. In a broader sense, this can be visualized as combining two groups of two objects each, leading to a total of four objects.